

Technical Exercise

MIT Sloan Finance Group – Technical Associate Position

This document describes the technical exercise for the MIT Sloan Finance Group research assistant position. Feel free to complete this exercise in the programming language in which you are most comfortable, but specific function hints are provided only in Python / Stata / R. The exercise asks you to explore the response of some core macroeconomic variables to changes in total factor productivity (TFP), following [Ramey \(2016\)](#). Specifically, you will use a local projection approach to compute the response of these variables at various horizons to “shocks” to TFP, using two different measures of the shock series.

Instructions. The exercise materials should be returned by email as a single folder with a README file indicating how to run the code and produce the plots. A compressed or zipped version of the folder is fine. The folder should be self-contained, containing all relevant data and code, so that a user could immediately run the code provided they had only a copy of Python/R/stata and the relevant modules/packages. The intended time of the exercise is three hours, but applicants can take more time to finish as needed. Any outputs that are generated (e.g., images, log files, etc) should be output into a single output subfolder. *Please also include these outputs with your submission.*

Data Download. The data for this assignment comes from four sources:

1. FORD TFP shocks: the file `ford_tfp.csv` contains the TFP shock series from [Francis, Owyang, Roush, and DiCecio \(2014\)](#). The “Quarter” notation is such that e.g., 1949.0 = 1949 Q1, 1949.25 = 1949 Q2, etc.
2. Fernald TFP shocks: download the file https://www.frbsf.org/economic-research/files/quarterly_tfp.xlsx and extract the variable `dtfp_util` from the quarterly sheet.

3. Shiller shock price data: download the file

http://www.econ.yale.edu/shiller/data/ie_data.xls

and extract the variable S&P Comp.P from the Data sheet.

4. Macroeconomic data: from the Federal Reserve data repository FRED (fred.stlouisfed.org), download the following series:

Code	Definition
CNP160V	Civilian Noninstitutional Population
GDP	Gross Domestic Product
GDPDEF	GDP Deflator
HOANBS	Hours
PCE	Consumption
PNFI	Non-Residential Investment

The use of an automated tool to download from FRED, such as `pandas-datareader` in Python / `fredr` in R / `import fred` in stata, is preferred.

Merge these datasets together into a single dataframe (i.e., using `pandas`) at quarterly frequency, taking means over variables to convert from monthly to quarterly frequency as necessary (the `resample` function in `pandas` may be useful here).

Data Processing. Please update the data according to the following steps:

1. For each variable in GDP, consumption, non-residential investment, and stock prices (S&P Comp.P from the Shiller data), compute a log real per-capita version by dividing through by population and the GDP deflator, and then taking logs. Multiply each series by 100 so that it is expressed in percent.
2. Compute log hours per capita by dividing hours by population, taking logs, and multiplying by 100.
3. Compute log real labor productivity by dividing GDP by hours and the GDP deflator, taking logs, and multiplying by 100.

Local Projection. The effects of the shocks, denoted ε , will be measured using local projection regressions following [Jordà \(2005\)](#). Specifically, for each variable z and each

horizon h , run the linear regression

$$z_{t+h} = \alpha_h + \sum_{j=0}^2 \theta_{h,j} \varepsilon_{t-j} + \gamma z_{t-1} + \phi_h' X_{t-1} + u_{t,t+h}.$$

In words, this is a regression of the dependent variable z , h periods ahead (z_{t+h}), on the current value of the shock and two lags, the lagged value of the dependent variable (z_{t-1}) and on a vector of lagged control variables. The term $u_{t,t+h}$ is the regression residual. Note that this regression must be run separately for each horizon of interest h . Because the residuals $u_{t,t+h}$ span overlapping time periods, they are likely to be serially correlated, so standard errors should be Newey-West corrected for h lags.¹

Please run two sets of local projections varying the horizon from 0 to 20Q, one set using the Fernald TFP shocks (`dtfp_util`) and the other using the FORD TFP shocks, as empirical proxies for $\{\varepsilon_{t-j}\}_{j=0}^2$, respectively.

The rest of the specification is as follows.

1. For the dependent variables, $\{y_{t+h}\}_{h=0}^{20}$, use log real GDP per capita, log labor productivity, log real stock prices per capita, log real consumption per capita, log non-residential investment per capita.
2. For the controls, X_{t-1} , include log real GDP per capita, log real stock prices per capita, log real labor productivity, and the dependent variable z_{t-1} (if it is not one of the control variables already listed).

Plots. Please report your results for each set of regressions in the form of a plot. The plot should contain subplots showing the estimated response of each dependent variable to the current shock ε_t , denoted $\theta_{h,0}$ in the regression, where the x-axis is the horizon h . The plots should also include 66.7% confidence bands around these estimates.²

Economic interpretation. Please write a short paragraph describing the economic interpretation of the estimated local projection impulse responses. What do the figures tell you about the relationship between TFP shocks and real GDP? (You can include this paragraph at the end of your README file.)

¹in Python `statsmodels`, this can be done using the 'HAC' option in `get_robustcov_results`. In R, one can use the `NeweyWest` function in the `sandwich` package. In Stata, one can use the `newey` function.

²In stata, you may find it helpful to use the `coefplot` package. Alternatively, if you struggle to make such a coefficient plot in stata, please output point estimates and standard errors to a well-formatted table (e.g., using `estout`).

Evaluation. The exercise will be evaluated according to the following criteria.

1. Correctness: whether the code correctly produces the output.
2. Readability: how readable the code is to an outside user. Good organization and appropriate use of comments will lead to a high score on this dimension.
3. Reusability: how re-usable the code would be for a different application. Using flexible programming, splitting out subroutines into separate functions for reuse, and avoiding hard-coding values, will lead to a high score on this dimension. Particularly relevant for this exercise will be the ability to run both sets of results with minimal coding changes and little to no copy-pasting.
4. Speed: it is preferred (but not required) to finish in three hours.

References

- FRANCIS, N., M. T. OWYANG, J. E. ROUSH, AND R. DICECIO (2014): “A Flexible Finite-Horizon Alternative to Long-Run Restrictions With an Application to Technology Shocks,” *Review of Economics and Statistics*, 96, 638–647.
- JORDÀ, Ò. (2005): “Estimation and Inference of Impulse Responses by Local Projections,” *American Economic Review*, 161–182.
- RAMEY, V. A. (2016): “Macroeconomic Shocks and Their Propagation,” *Handbook of Macroeconomics*, 2, 71–162.